LiteArm

DH Parameters

NEXFORM ROBOTICS

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Revision history

Version	Date	Change
V1.0	2026-11-02	First public release for customers

Contents

1 Coordinate frames and sign conventions 1

2 Modified DH parameter table 1

1 Coordinate frames and sign conventions

This document uses the modified DH (Craig / MDH) convention. The homogeneous transform between adjacent link frames is:

$$T_{i-1\to i} = R_x(\alpha_i) \cdot T_x(a_i) \cdot R_z(\theta_i) \cdot T_z(d_i)$$

$$\theta_i = \text{offset}_i + s_i \cdot q_i$$

where i is the joint number (1–7); α_i is the twist about the x axis; a_i is the link length along x; d_i is the offset along z; offset_i is the zero-position offset; s_i is the joint direction sign; and q_i is the joint angle. Angles are in degrees, lengths in metres, and q in radians.
Note that the modified DH transform order is $R_x \to T_x \to R_z \to T_z$, whereas standard DH is $R_z \to T_z \to T_x \to R_x$. Do not mix the two.

2 Modified DH parameter table

Table 1 LiteArm modified DH parameters

Joint i	α_i (°)	a_i (m)	d_i (m)	θ offset (°)	s_i
1	0	0	0.13325	0	+1
2	−90	0	0	0	+1
3	90	0	0.2995	0	+1
4	−90	0.038	0	0	−1
5	90	−0.038	0.2225	0	+1
6	−90	0	0	90	−1
7	90	−0.074	0	0	−1

All seven joints are revolute. $s_i = +1$ means the positive joint direction agrees with the DH z axis, and −1 means it is opposite. In the table, α and the offsets are exact multiples of ±90°, while a and d are design values.
For kinematic calculations, substitute the parameters for each joint number into the transform above. Dynamic parameters for each link are provided separately.